MATHEMATICS
University of NORTH ALABAMA Spring 2025 Math

Problem of the Week: Problem #3

Due noon Monday, February 3Open to all UNA undergraduates

Directions: Deliver your written solution to me (Dr. Bryant) in CPTMTH 328 by noon Monday, February 3 or, if you cannot find me, you may leave it with Ms. Buckley in CPTMTH 377. All work must be your own. You are required to justify your response fully. If you utilize a well-known theorem, state the theorem's full name and proposition, showing that the conditions of the theorem are met. Good luck!

Problem #3: Let $a_1, a_2, \ldots, a_{2025}$ be positive integers such that

- $a_1, a_2, \ldots, a_{2025}$ is a permutation of $1, 2, \ldots, 2025$ and
- $|a_1 - a_2|, |a_2 - a_3|, \ldots, |a_{2024} - a_{2025}|$ is a permutation of $1, 2, \ldots, 2024$.

Prove that $\max\{a_1, a_{2025}\} \geq 507$.